Peizhi Liu

2233 Tech Drive
Mudd Hall, Room 3302
Evanston, IL 60208

peizhi.liu@u.northwestern.edu
+1 (248) 824-0359
peizhiliu.com

RESEARCH INTERESTS

I am a third-year Ph.D. candidate at Northwestern University under Professor Peter Dinda. I am broadly interested in parallel and distributed computing systems. My current research is focused on enhancing the communication and performance of these systems with the use of novel models, mappings, and mechanisms. I am an NSF GRFP Fellow.

EDUCATION

Northwestern University, Evanston, IL

Ph.D. Candidate, Computer Science	Feb 2026 – *Present*
Ph.D. Student, Computer Science	Mar 2024 – Feb 2026
M.S. Student, Computer Science	Sep 2022 – Jun 2023
B.S. Student, Computer Science (Honors)	Sep 2019 – Jun 2023

RESEARCH EXPERIENCE

Graduate Student, Northwestern University — Mar 2024 – Oct 2025

Parameterized Algorithms and Parameter Selection for Fast GPU-GPU Collective Communication **[1]**
Mentors: Professor Peter Dinda and Dr. Michael Wilkins

- Developed parameterized collective algorithms for GPU-GPU communication that are flexible and performant on modern supercomputer topologies. Used a simulation-based approach to choose suitable collective parameters. Received $25,000 in AWS resources and a total of 60,000 GPU Hours of compute time across three supercomputers through the National Artificial Intelligence Research Resource (NAIRR) Pilot.
- Research Question – Can parameterized algorithms on GPUs be used to achieve performant collective communication without sacrificing portability on modern supercomputers?

Frequent Subgraph Detection in Vector Dataflow Graphs for Village
Mentor: Professor Peter Dinda

- Worked on analyzing program dataflow graphs from Village, a new implementation of the NESL high-level programming language, to extract frequent and expensive subgraphs for compiler tiling optimizations. The discovered subgraphs can potentially be used to guide hardware design decisions for accelerator types.
- Research Question - From a hardware design perspective, do common scientific applications share computational kernels that manifest as expensive subgraphs in Village?

Enabling NISQ Chiplet Architectures via an Entanglement Distribution Network
Mentor: Professor Kate Smith

- Developed a discrete-event simulation framework built on top of Cirq to evaluate entanglement distribution network (EDN) topologies, communication protocols, routing algorithms, and qubit fidelity. Implementing a custom network-routing transpiler pass in Qiskit to evaluate the EDN's impact on applications.
- Research Question - Can heterogeneous design in qubit noise profiles enable better fidelity circuits on chiplet-based NISQ architectures via a quantum entanglement network?

Undergraduate Student, Northwestern University — Sep 2019 – Jun 2023

Generalized Collective Algorithms for the Exascale Era **[2]**
Mentors: Professor Peter Dinda and Dr. Michael Wilkins

- Developed parameterized versions of MPI All-Gather, All-Reduce, and Bcast collective algorithms to enable machine learning-based autotuning performance enhancements that can potentially account for dynamic, non-programmatic factors of a supercomputer.
- Achieved up to 4.5x speedup compared to original MPICH and native Cray MPICH implementations on the Frontier exascale supercomputer.

PDNS: A Fully Privacy-Preserving DNS **[3,4]**
Mentors: Professor Aleksandar Kuzmanovic and Dr. Yunming Xiao

- Designed and implemented a privacy-preserving DNS recursive resolver by leveraging single-server private information retrieval (PIR). This allows users to retrieve DNS records anonymously without the server knowing.
- Also developed and ran benchmarks for client and server-side query performance, including user webpage performance tests. Determined that PDNS offers significant privacy guarantees with acceptable performance.

Variation in Derived Extrasolar Planet Spectrum
Mentors: Professor Melville Ulmer and Dr. Cobi Rabinowitz

- Developed an end-to-end spectrum extraction pipeline using the CHARIS Data Reduction pipeline and pyKLIP to explore variability in exoplanet spectra for HD 1160 b and Kappa And b. These techniques have the potential to be used for detecting solar flare events and the exoplanets' rotational velocity.
- Verified previously published Kappa And b spectrum and discovered substantial variability in the spectrum, either due to physical, instrumental, or pipeline processing. Presented findings to other ISG students.

INDUSTRY EXPERIENCE

Software Development Engineer Intern, Amazon Jun 2022 – Sep 2022
Extending Log Filters and Cache for Recommendations Service Testing
Hosts: Christian Tan, Zimeng Zhang

- Worked in the Personalization organization to implement extensible test log filtering with DynamoDB to easily onboard new recommendations services without code changes. Addressed problems with insufficient service test data by designing a test data caching scheme with S3.
- Used build and pipeline systems to deploy changes onto production servers and to monitor service state for potential issues and bugs.

Junior Intern (Software Engineering), Robert Bosch LLC Jun 2021 – Aug 2021
Integrating Protocol Buffers in MCU Communications
Hosts: Syed Ahson, Barrington Brown

- Worked in the Perfectly Keyless team to integrate Protocol Buffers to be used for MCU cryptographic handshakes.
- Analyzed stack and MCU RAM usage of the protocol library to inform the customer of potential decision tradeoffs.

SELECTED PROJECTS

Accelerated Journal Bearing Hydrodynamic Lubrication Simulation on CUDA **(Undergraduate Thesis)**
Mentors: Professor Nikos Hardavellas and Professor Peter Dinda

- Developed a GPU-accelerated solution for modeling the tribology of journal bearings under hydrodynamic and non-Newtonian lubrication. GPU-specific optimizations include tiling, reduction, memory coalescing, and bank conflict avoidance.
- Achieved over 9x speed-up in total simulation time over a comparable sequential CPU simulation for medium-sized meshes.

Benchmarking Overhead of Running Neural Networks in OP-TEE
Mentor: Professor Peter Dinda

- Investigated memory and performance limitations of deep learning in the Open Portable Trusted Execution Environment (OP-TEE), an implementation of Arm TrustZone running on an embedded device (Raspberry Pi 3b). Specifically focused on a layer-based partitioning technique to reduce the model's memory footprint.
- Discovered layer-based partitioning increases the maximum memory usage while only marginally decreasing average memory usage. Further experiments comparing the use of Secure Storage versus encrypted Shared Memory revealed significant performance improvements of the latter over the former.

PUBLICATIONS / DEMOS

[1] Liu, Peizhi, Sean Rhee, Michael Wilkins, and Peter Dinda. "Parameterized Algorithms and Parameter Selection for Fast GPU-GPU Collective Communication." In *2025 33rd International Symposium on Modeling, Analysis and Simulation of Computer and Telecommunication Systems (MASCOTS)*, pp. 1-9. IEEE, 2025.

[2] Wilkins, Michael, Hanming Wang, Peizhi Liu, Bangyen Pham, Yanfei Guo, Rajeev Thakur, Peter Dinda, and Nikos Hardavellas. "Generalized Collective Algorithms for the Exascale Era." In *2023 IEEE International Conference on Cluster Computing (CLUSTER)*, pp. 60-71. IEEE, 2023.

[3] Xiao, Yunming, Chenkai Weng, Ruijie Yu, Peizhi Liu, Matteo Varvello, and Aleksandar Kuzmanovic. "PDNS: A Fully Privacy-Preserving DNS." In *Proceedings of the ACM SIGCOMM 2023 Conference*, pp. 1182-1184. 2023.

[4] Xiao, Yunming, Peizhi Liu, Ruijie Yu, Chenkai Weng, Matteo Varvello, and Aleksandar Kuzmanovic. "Collusion Resistant DNS With Private Information Retrieval." *arXiv preprint arXiv:2507.20806* (2025).

TEACHING AND MENTORING EXPERIENCE

Teaching

Peer Mentor (Undergraduate TA), Northwestern University

Winter 2023	CS 336 – Design & Analysis of Algorithms	with Instructor Donald M. Stull
Winter 2022	CS 343 – Operating Systems	with Prof. Peter Dinda
Fall 2021	CS 336 – Design & Analysis of Algorithms	with Prof. Konstantin Makarychev

Teacher for AP Statistics Course

Taught weekly AP Statistics classes to a class of 11 high school students over the summer in 2021. Prepared slides, class material, and graded weekly homework assignments.

Teacher for Python Programming Course

Taught elementary and middle school students Python Programming as part of my church's summer camp program in 2023. For more advanced students, I also taught basic web development with HTML, CSS, and Javascript.

Mentoring

- Mentoring two undergraduate students on current research project.
- Mentored Sean Rhee (undergraduate student at Northwestern University, now Ph.D. student at NYU) on the *Parameterized Algorithms and Parameter Selection for Fast GPU-GPU Collective Communication* project.

SERVICES

Co-Reviewer for SC 2026
Co-Reviewer for HPDC 2026
Co-Reviewer for IPDPS 2026
Co-Reviewer for HPDC 2025

SKILLS

Languages: C/C++, Python, CUDA, Rust, Java, JavaScript, Bash
Technologies: MPI, NCCL, RDMA, Infiniband, PyTorch, ARM TrustZone (OP-TEE), Protocol Buffers
Tools: Unix/Linux, GDB, Git, AWS, Firebase, Jupyter Notebook

AWARDS AND HONORS

NSF Graduate Research Fellowship Program (GRFP) 2025 – 2028
NAIRR Pilot Resource Award 2024 – 2026
- $25K in AWS resources, 60,000 GPU Hours across three HPCs

Dean's List, all quarters 2019 – 2023
National Merit Scholarship 2019 – 2023
Illinois Space Grant 2020
- Summer research experience with Prof. Melville Ulmer

Grand Award Winner at the Science and Engineering Fair of Metro Detroit 2019
- Participated in Intel ISEF 2019